

E-Commerce Cart Abandonment & Conversion Optimization

1. Project Overview

An e-commerce company is receiving a large number of website visitors and product interactions, but a significant number of users are not completing their purchases. Management wants to understand where customers drop out of the purchase funnel, identify the factors associated with abandonment, and find opportunities to improve conversion and revenue.

2. Dataset Summary

Customers - (5,000) Customer demographics & segments

Products - (300) Product, category, price, cost & stock

Sessions - (30,000) Main conversion funnel data

Orders - (4,558) Purchases, revenue, discounts & payment

Delivery - (4,558) Delivery performance

Returns - (486)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
#import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3
```

```
#load datasets in jupyter notebook
df_cutomers = pd.read_csv('Customers.csv')
df_delivery = pd.read_csv('Delivery.csv')
df_orders = pd.read_csv('Orders.csv')
df_products = pd.read_csv('Products.csv')
df_returns = pd.read_csv('Returns.csv')
df_sessions = pd.read_csv('Sessions.csv')
```

```
df_cutomers.shape

(5000, 7)

#check their structure
df_cutomers.info()

<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Customer_ID      5000 non-null   str
1   Age              5000 non-null   int64
2   Gender           5000 non-null   str
3   City             5000 non-null   str
4   State            5000 non-null   str
5   Customer_Type     5000 non-null   str
6   Signup_Date       5000 non-null   str
dtypes: int64(1), str(6)
memory usage: 273.6 KB
```

```
# check their summary statistics
df_cutomers.describe(include= 'all')
```

	Customer_ID	Age	Gender	City	State	Customer_Type	Signup_Date
count	5000	5000.000000	5000	5000	5000	5000	5000
unique	5000	NaN	3	38	15	2	898
top	CUST100001	NaN	Male	New Delhi	Maharashtra	Returning	2024-12-08
freq	1	NaN	2490	382	763	3189	15
mean	NaN	38.855800	NaN	NaN	NaN	NaN	NaN
std	NaN	12.246036	NaN	NaN	NaN	NaN	NaN
min	NaN	18.000000	NaN	NaN	NaN	NaN	NaN
25%	NaN	29.000000	NaN	NaN	NaN	NaN	NaN
50%	NaN	39.000000	NaN	NaN	NaN	NaN	NaN
75%	NaN	49.000000	NaN	NaN	NaN	NaN	NaN
max	NaN	60.000000	NaN	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values
- **Feature Engineering:** ◦ Created **age_group** column by binning customer ages.
- **change datatype - (signup_date):** Changed datatype str to datetime datatype.
- **Database Integration:** Connected Python script to SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in SQL Server to answer key business questions:

- **Funnel analysis**
1. **overall conversion rate** – Out of all sessions/visitors, how many finally purchased?

```
25  --1. What is the overall conversion rate?
26
27  ∨ SELECT COUNT(DISTINCT
28  CASE WHEN [Purchase] = 'Yes' THEN [Session_ID]
29  END) * 100.0
30  / NULLIF(COUNT(DISTINCT[Session_ID] ), 0) AS Conversion_Rate
31  FROM [dbo].[Sessions];
```

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Results Messages

Conversion_Rate
15.193333333333

2. customer drop-off– Identified where higher customers drop off ‘

Product View → Cart

Cart → Checkout

Checkout → Payment

Payment → Purchase

```
5
6 --1.Product View → Cart
7 SELECT (Product_Views - Add_To_Cart_Count) * 100.0
8       / NULLIF(Product_Views, 0) AS View_To_Cart_Dropoff
9 FROM (SELECT
10       SUM([Product_Views]) AS Product_Views,
11       COUNT(CASE WHEN [Add_To_Cart] = 'Yes' THEN 1 END)
12       AS Add_To_Cart_Count
13 FROM [dbo].[Sessions] ) AS A;
```

Results	Messages
View_To_Cart_Dropoff	
91.737813731590956	

```
--2.Cart → Checkout
select (Cart_Count - Checkout_count) *100.0 /
nullif(Cart_Count , 0) as Cart_to_Checkout_dropoff
from (select
count(case when [Add_To_Cart] = 'Yes' then 1 end)
as Cart_Count,
COUNT(CASE WHEN [Checkout_Started] = 'Yes' THEN 1 END)
AS Checkout_count
from [dbo].[Sessions]) B;
```

Results	Messages
Cart_to_Checkout_dropoff	
32.858003615183	

```
55
56 --3.Checkout → Payment
57
58 select ( Checkout_count - Payment_count) *100.0 /
59 nullif(Checkout_count , 0) as Checkout_to_Payment_dropoff
60 from (select
61 count(case when [Checkout_Started] = 'Yes' then 1 end)
62 as Checkout_count,
63 COUNT(CASE WHEN [Payment_Attempt] = 'Yes' THEN 1 END)
64 AS Payment_count
65 from [dbo].[Sessions]) B;
```

Results	Messages
Checkout_to_Payment_dropoff	
13.909661980257	

```
66
67 --4.Payment → Purchase
68
69 select ( Payment_count - purchase_count) *100.0 /
70 nullif(Payment_count , 0) as Payment_to_purchase_dropoff
71 from (select
72 count(case when [Payment_Attempt] = 'Yes' then 1 end) as Payment_count,
73 COUNT(CASE WHEN [Purchase] = 'Yes' THEN 1 END) AS purchase_count
74 from [dbo].[Sessions]) B;
```

Results	Messages
Payment_to_purchase_dropoff	
20.813064628214	

3. cart abandonment rate: Customers who added a product to cart but did not purchase.

```
75
76 ---3. What is the cart abandonment rate?
77
78 SELECT COUNT(CASE WHEN Add_To_Cart = 'Yes'
79 AND Purchase = 'No' THEN 1 END) * 100.0
80 / NULLIF(COUNT(CASE WHEN Add_To_Cart = 'Yes' THEN 1 END),0 ) AS Cart_Abandonment_Rate
81 FROM Sessions;
82
83
84 ---4. What is the checkout abandonment rate?
85
86 SELECT COUNT(CASE WHEN [Checkout_Started] = 'Yes'
87 AND Purchase = 'No' THEN 1 END) * 100.0
88 / NULLIF(COUNT(CASE WHEN [Checkout_Started] = 'Yes' THEN 1 END),0 ) AS Checkout_Abandonment_Rate
89 FROM Sessions;
90
91
92 ---5. What is the payment failure rate?
93
94 SELECT COUNT(CASE WHEN [Payment_Success] = 'NO' THEN 1 END) * 100.0
95 / NULLIF(COUNT(*), 0) AS Payment_Failure_Rate
96 FROM [dbo].[Sessions];
97
98 ----[Customer]
99
100 ---6. What is the cart abandonment rate for new vs returning customers?
```

Results	Messages
Cart_Abandonment_Rate	
54.227756577626	
Checkout_Abandonment_Rate	
31.827699670954	
Payment_Failure_Rate	
82.080000000000	

4. **checkout abandonment rate**– Customers who Checkout but did not purchase.
5. **payment failure rate:**
6. **Top 10 customer generate the most revenue:**

```

26
27  ---8. Which customers generate the most revenue?
28
29  SELECT top 10
30  c.[Customer_ID],
31  SUM(o.[Revenue]) AS Total_Revenue,
32  COUNT(DISTINCT o.[Order_ID]) AS Total_Orders
33  FROM [dbo].[Customers] as c
34  join [dbo].[Orders] as o
35  ON c.[Customer_ID] = o.[Customer_ID]
36  GROUP BY
37  c.[Customer_ID]
38  ORDER BY Total_Revenue DESC;
39

```

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Results	Messages			
Customer_ID	Total_Revenue	Total_Orders		
CUST100875	275109.45	5		
CUST102922	263889.65	2		
CUST104856	257356.4	3		
CUST104593	246915	4		
CUST102538	244566.6	2		
CUST103694	244242.4	4		
CUST103179	240589.75	4		
CUST103602	237478.9	3		
CUST103633	215330.8	2		
CUST100946	202098.6	3		

7. **Top 10 Products high views but low purchases:**

89 %

▼

✖

2

⚠

0

↑

↓

Results

Messages

	Product_ID	Product_Name	total_views	total_Purchase	Conversion_Rate
1	P10097	Nova Kids Item 97	524	24	4.580152671755725190839
2	P10004	Zenith Footwear Item 4	517	16	3.094777562862669245647
3	P10162	DailyBuy Fragrance Item 162	510	25	4.901960784313725490196
4	P10230	StyleHub Footwear Item 230	506	18	3.557312252964426877470
5	P10102	StyleHub Fragrance Item 102	505	17	3.366336633663366336633
6	P10236	Vistara Staples Item 236	499	26	5.210420841683366733466
7	P10029	FitMax Accessories Item 29	496	18	3.629032258064516129032
8	P10141	Zenith Kitchen Item 141	492	17	3.455284552845528455284
9	P10159	DailyBuy Haircare Item 159	491	15	3.054989816700610997963
10	P10160	Nova Staples Item 160	489	16	3.271983640081799591002

8. Is mobile conversion lower than desktop:

```

188  ---12. Is mobile conversion lower than desktop?
189
190  SELECT [Device],
191  COUNT(DISTINCT [Session_ID]) AS Total_Sessions,
192  COUNT(DISTINCT CASE WHEN [Purchase] = 'Yes' THEN [Session_ID] END) AS Purchases,
193  COUNT(DISTINCT CASE WHEN [Purchase] = 'Yes' THEN [Session_ID] END) * 100.0
194  / NULLIF(COUNT(DISTINCT [Session_ID]), 0) AS Conversion_Rate
195  FROM [dbo].[Sessions]
196  GROUP BY [Device]
197  ORDER BY Conversion_Rate DESC;
198

```

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Results Messages

	Device	Total_Sessions	Purchases	Conversion_Rate
1	Desktop	8116	1243	15.315426318383
2	Mobile	20469	3105	15.169280375201
3	Tablet	1415	210	14.840989399293

9. conversion rate vary by estimated delivery time :

```

231
232
233  ----[Delivery]
234
235  ---16. How does conversion rate vary by estimated delivery time ?
236
237  SELECT
238  [Delivery_Estimated_Days],
239  COUNT(DISTINCT [Session_ID]) AS Total_Sessions,
240  COUNT(DISTINCT CASE WHEN [Purchase] = 'Yes' THEN [Session_ID] END) AS
241  COUNT(DISTINCT CASE WHEN [Purchase] = 'Yes' THEN [Session_ID] END) *
242  / NULLIF(COUNT(DISTINCT [Session_ID]), 0) AS Conversion_Rate
243  FROM [dbo].[Sessions]
244  GROUP BY [Delivery_Estimated_Days]
245  ORDER BY [Delivery_Estimated_Days];
246

```

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Results Messages

	Delivery_Estimated_Days	Total_Sessions	Purchases	Conversion_Rate
1	2376	365	15.361952861952	
2	3648	570	15.625000000000	
3	5273	807	15.304380807889	
4	5399	869	16.095573254306	
5	4885	757	15.496417604912	
6	3587	599	16.699191524951	
7	2406	299	12.427265170407	
8	1534	180	11.734028683181	
10	892	112	12.556053811659	

10. conversion rate vary by estimated delivery time :

```

276 ---19. Which category has the highest return rate?
277
278 select p.[Category],
279 count(distinct o.[Order_ID]) as total_orders,
280 count(distinct r.[Return_ID]) as total_return_orders,
281 count(distinct r.[Return_ID]) *100.0/ nullif(count(distinct o.[Order_ID]),0) as return_rate
282 from [dbo].[Orders] as o
283 join Products p
284 on o.[Product_ID] = p.[Product_ID]
285 left join [dbo].[Returns] as r
286 on o.[Order_ID] = r.[Order_ID]
287 group by p.[Category]
288 order by return_rate desc;
289 --ans: fashion is the highest return rate (11.81%).

```

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Results Messages

Category	total_orders	total_return_orders	return_rate
Fashion	1143	135	11.811023622047
Grocery	730	86	11.780821917808
Home & Kitchen	666	72	10.810810810810
Beauty	508	52	10.236220472440
Electronics	1102	108	9.800362976406
Sports	409	33	8.068459657701

5.Data Modeling

Fact tables

Fact_Sessions

Fact_Orders

Dimension tables

Dim_Customer

Dim_Product

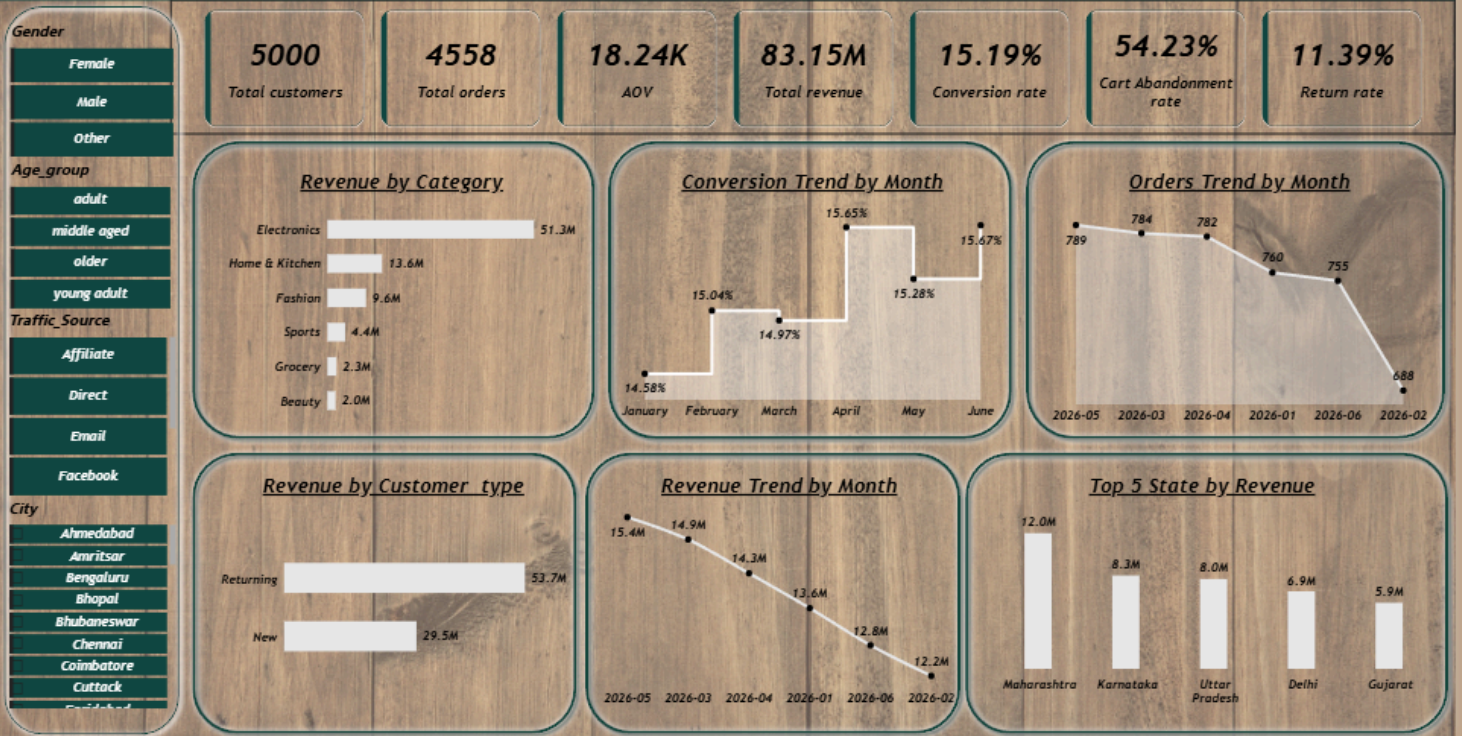
Dim_Delivery

Dim_return



6.Build Power BI Dashboard

Executive Overview



Conversion Funnel & Drop-off Analysis

